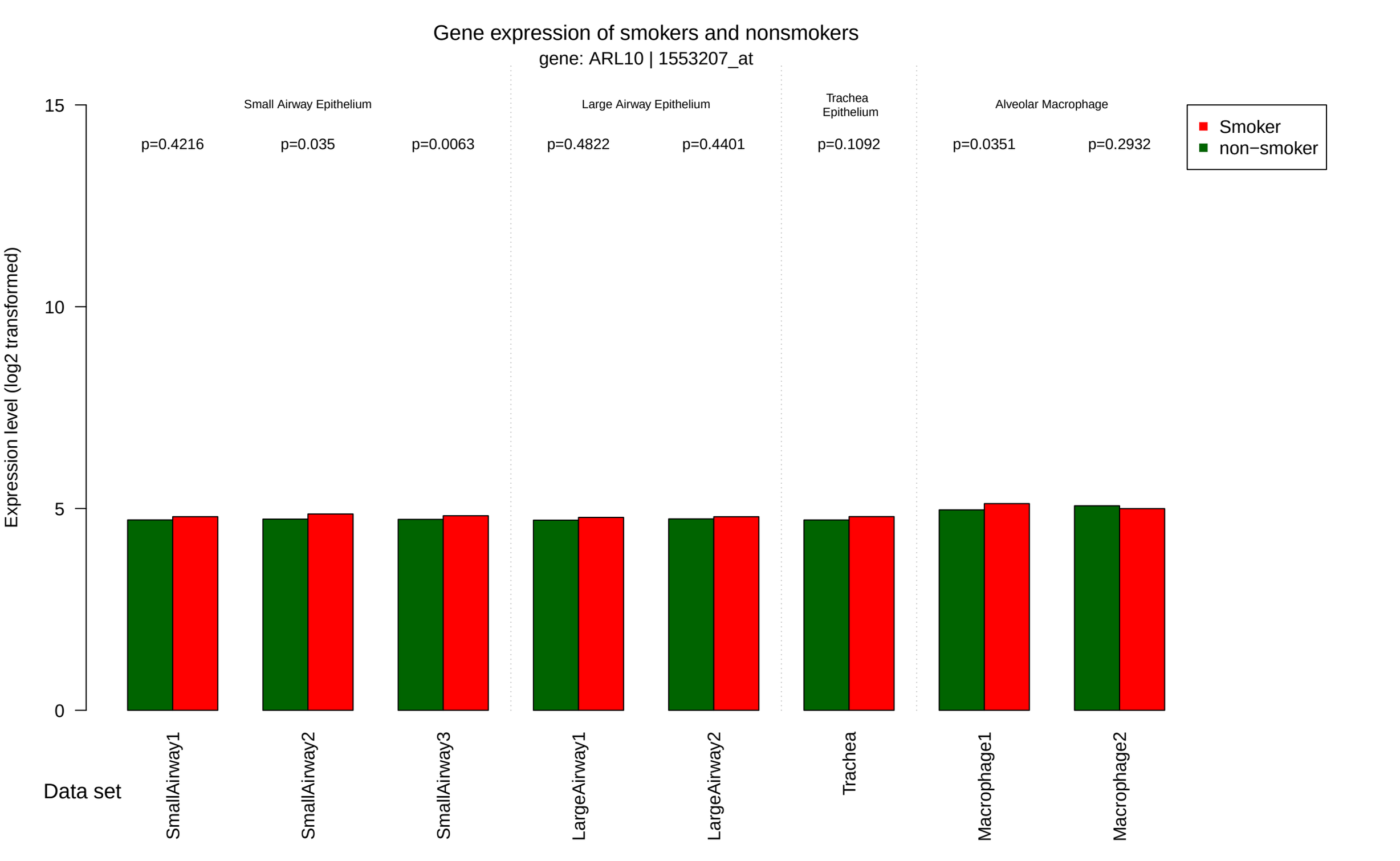
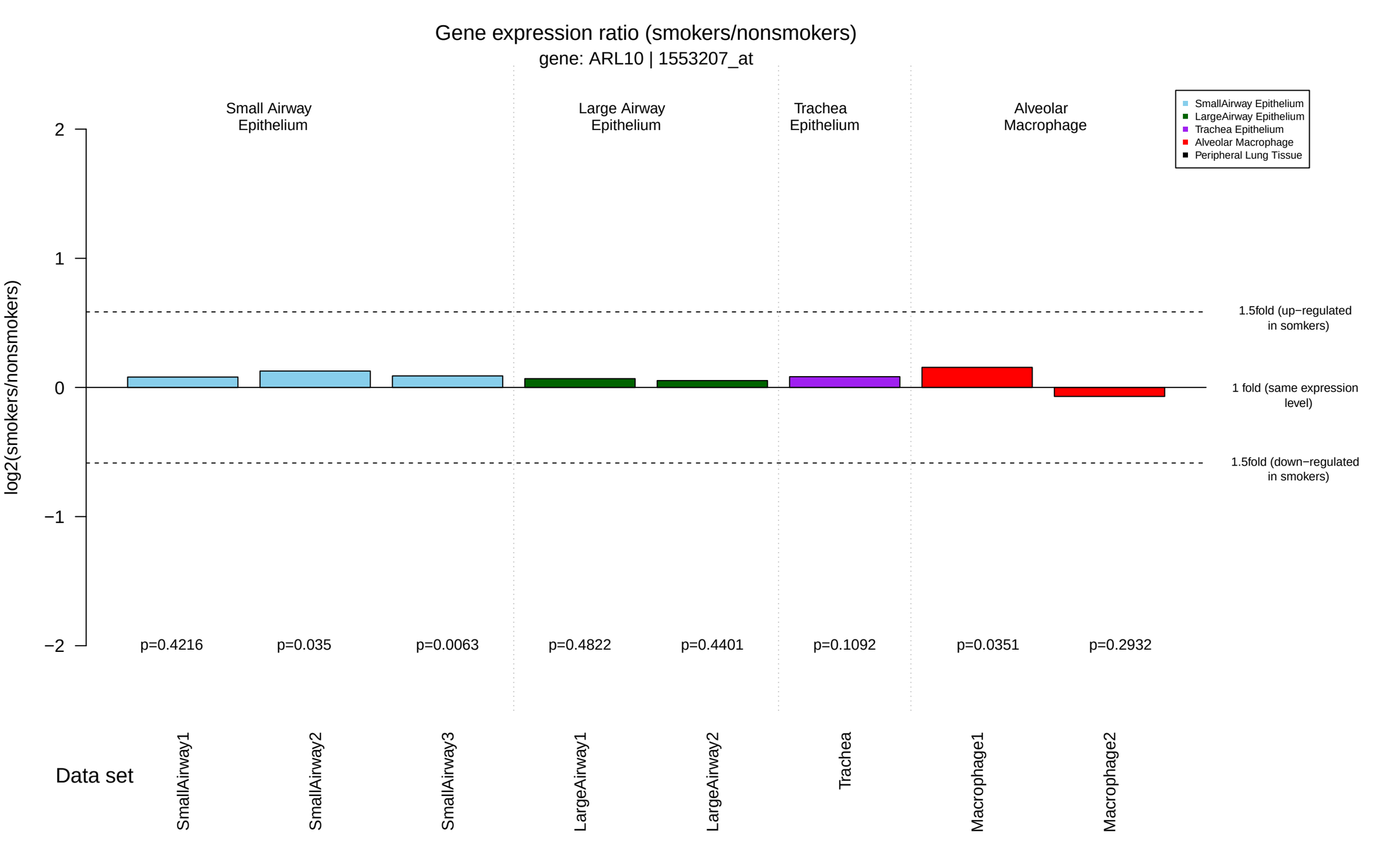
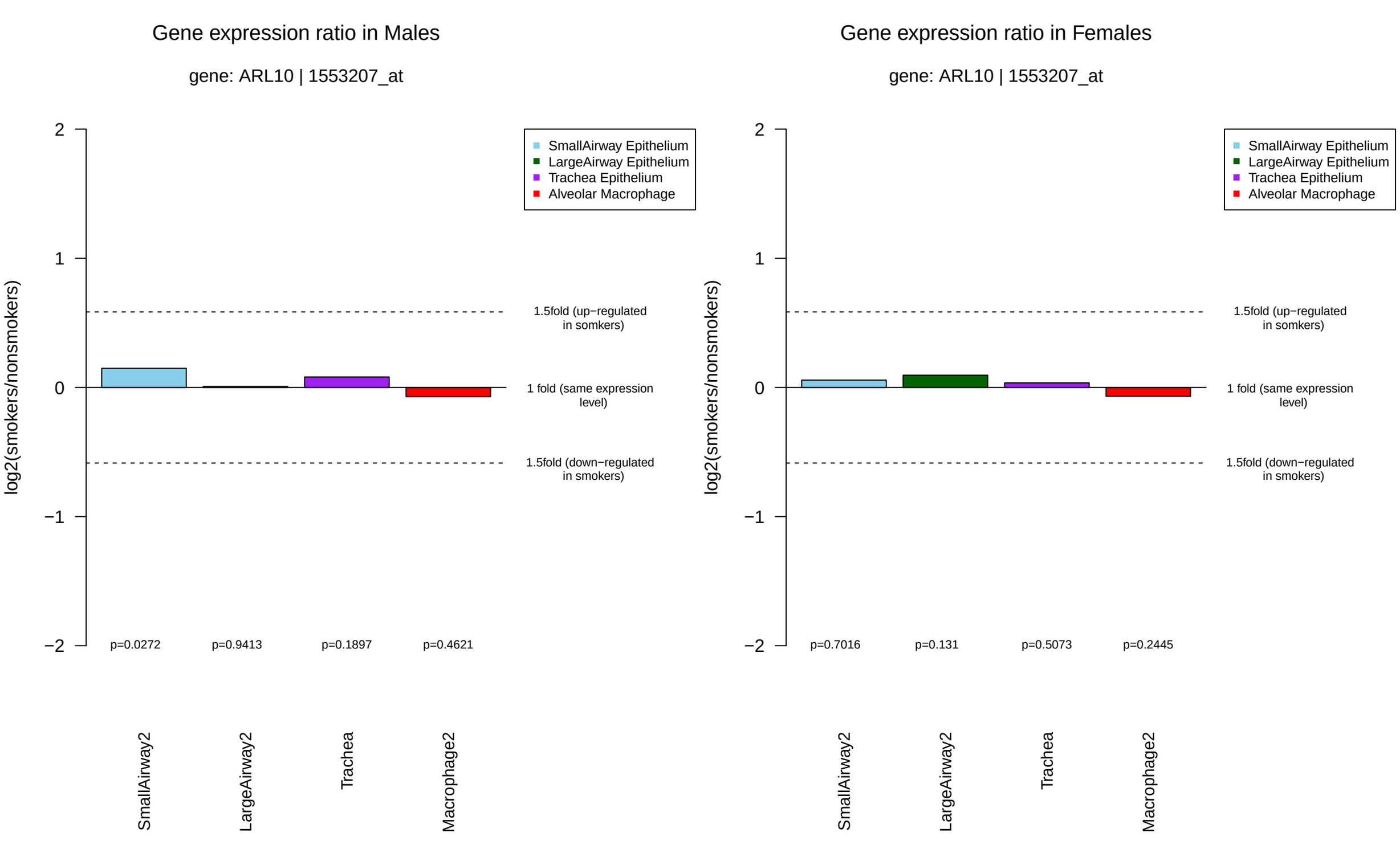
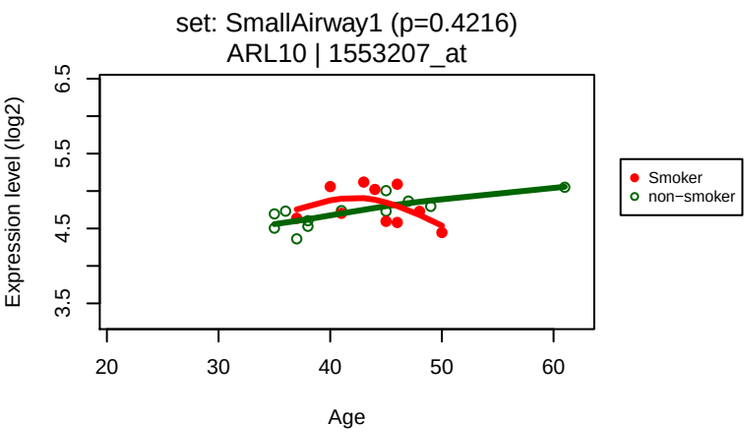
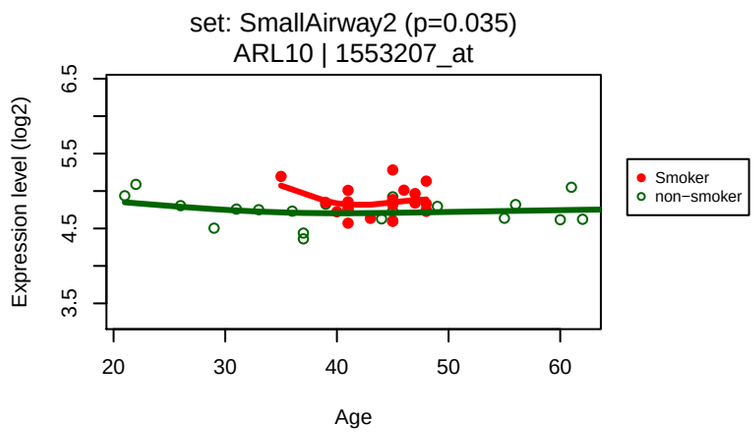
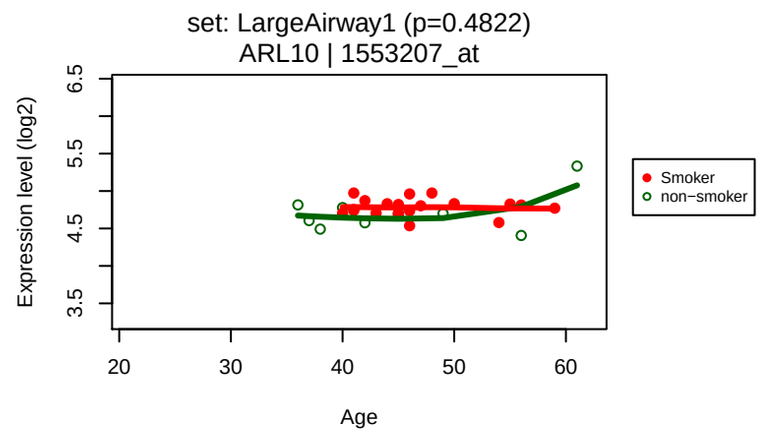

Gene expression of smokers and nonsmokers
gene: ARL10 | 1553207_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4216
p=0.035
p=0.0063
p=0.4822
p=0.4401
p=0.1092
p=0.0351
p=0.2932
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

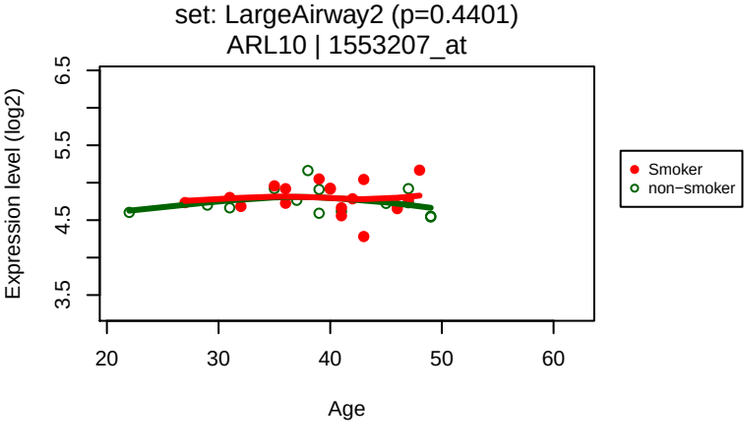

Gene expression ratio (smokers/nonsmokers)
gene: ARL10 | 1553207_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4216
p=0.035
p=0.0063
p=0.4822
p=0.4401
p=0.1092
p=0.0351
p=0.2932
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

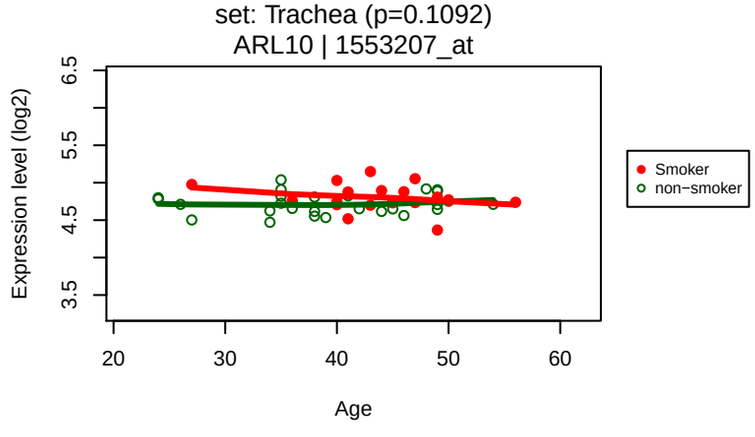

Gene expression ratio in Males
gene: ARL10 | 1553207_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0272
p=0.9413
p=0.1897
p=0.4621
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARL10 | 1553207_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7016
p=0.131
p=0.5073
p=0.2445
SmallAirway2
LargeAirway2
Trachea
Macrophage2

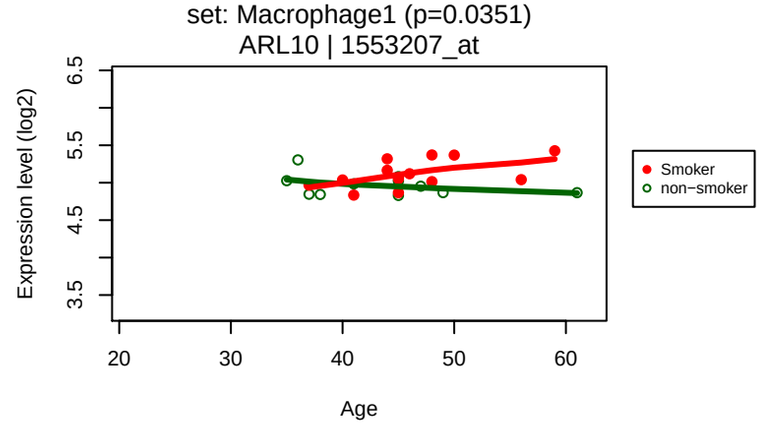

set: SmallAirway1 (p=0.4216)
ARL10 | 1553207_at
Expression level (log2)
Age
Smoker
non−smoker

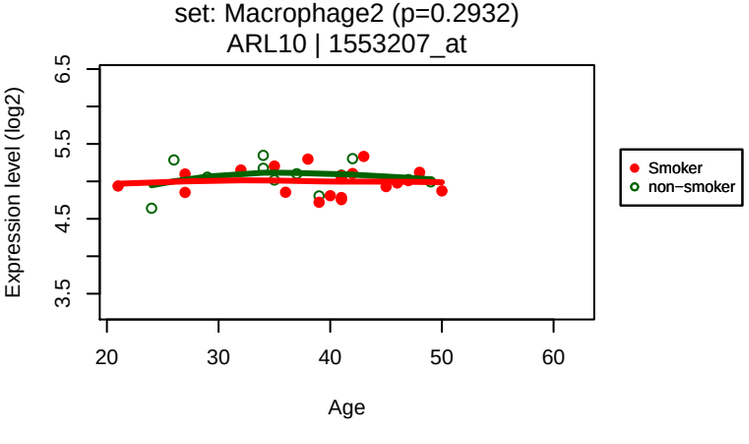

set: SmallAirway2 (p=0.035)
ARL10 | 1553207_at
Expression level (log2)
Age
Smoker
non−smoker

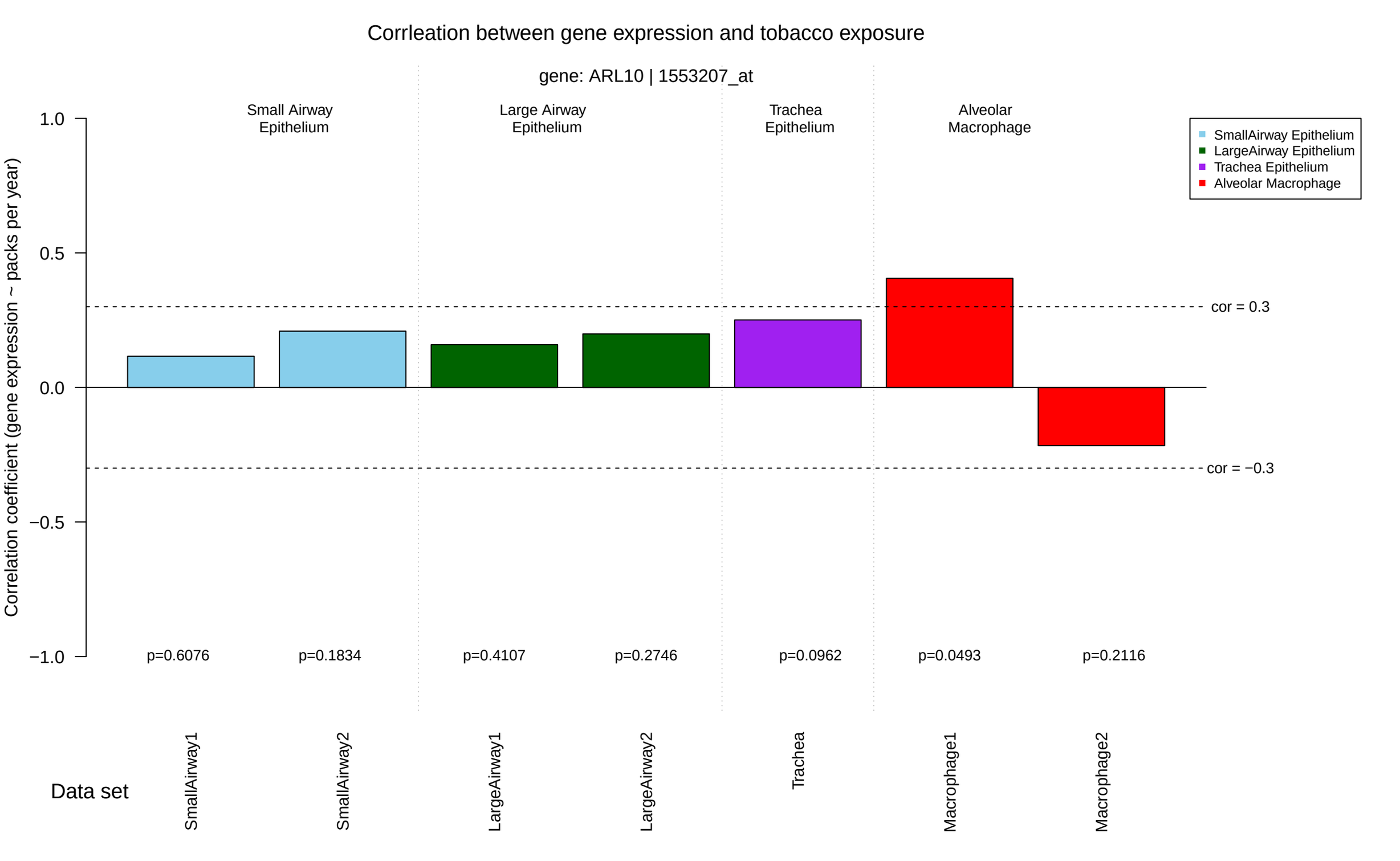

set: LargeAirway1 (p=0.4822)
ARL10 | 1553207_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4401)
ARL10 | 1553207_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1092)
ARL10 | 1553207_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0351)
ARL10 | 1553207_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2932)
ARL10 | 1553207_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARL10 | 1553207_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6076
p=0.1834
p=0.4107
p=0.2746
p=0.0962
p=0.0493
p=0.2116
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2